BRUNK INDUSTRIES, INC.
OPS SCORE FOR PER PART

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Explaining the OPS Formula

What is the OPS Formula and what does it do

A composite score per part to evaluate the performance and quality of manufactured parts

Each data point used is weighted by a factor to define level of importance to that data

When coupled with sales dollars, it can be used to define high and low return parts.

The larger the score, the more issues and lost opportunities

Process to Obtain the OPS Formula

Building a formula to allow us to analyze the parts and their fields

(SC x (1.5(S) + 5000(R) + 100(NC))) + (71.185 x (SH + Q + I + SU))/Revenue or Manufaturing

Scrap	The total amount of scrap that a product creates every time it is produced
Sorting Hours	The total amount of hours the product goes through sorting in the facility
RMA Count	The total amount of times this product has been returned
Non Conformance Count	The total number of products that did not conform to regulations
Standard Cost Per Unit	How much it costs to produce each part
Quality Time	The total amount of time the part spends in quality inspection
Inspection Time	The total amount of time the part spends in inspection
Setup Time	The total amount of time the part takes to setup
Sale Price	How much Brunk sells the part for

Validating the OPS Formula

We must test if the formula fairly reflects part-level performance across different use cases.

Part Name	Part Number	Our OPS / Manufacturing	Our OPS / Revenue	Brunk's OPS Per Part	Brunk's OPS Rev %
BD	X1944AAL	0.000878892	0	0.00003917	0.00871
Ethicon Trigger Assembly	D45681G01	0.25029795	0.168307424	0.0017	0.0012
Insulet Mech Rail	41082	0.026920034	0.204614715	0.265634708	0.201905
Insulet Battery Clip	13803	0.004824942	0.132424817	0.004030202	0.11061
Ethicon (1010) 420 Jaw	D01010P25	1.056737321	0.155757449	0.06864	0.00979

Situation Complication Question

What are our SCQ's?

Situation

We tested our custom OPS formula alongside Brunk's existing metric on five critical parts to assess alignment and effectiveness in identifying operational risk.

Complication

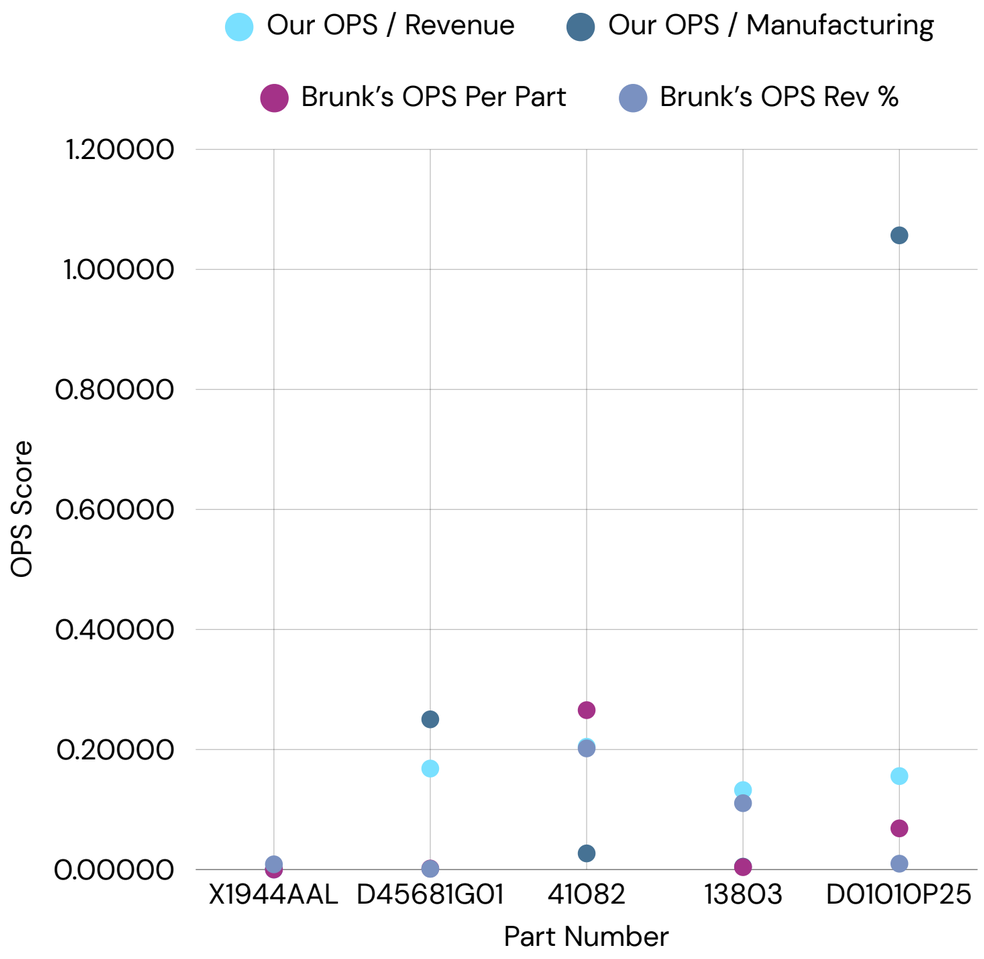
The two approaches produced different rankings and magnitudes, revealing gaps in how each formula interprets cost drivers like scrap, returns, and revenue impact.

Resolution

Which formula more accurately highlights high impact parts, and how can we refine the OPS model to better reflect real operational and financial performance?

Testing the formula for accuracy

Testing our OPS Formula compared to Brunk's OPS Formula for accuracy



High Variance in Rankings Between Formulas:

- Insulet Mech Rail ranks very high using Brunk's OPS Per Part (0.2656) and OPS Rev % (0.2019) but shows low manufacturing impact (0.0269) in your formula.
- Ethicon (1010) 420 Jaw is the highest in your OPS / Manufacturing (1.0567) but ranks much lower in Brunk's formula (0.06864).

Our Formula Highlights Operational Burden More Clearly:

- Parts like the Ethicon Trigger Assembly and 420 Jaw score significantly higher in your manufacturing and revenue-based OPS, suggesting strong signals for scrap, rework, or quality time.
- Brunk's values, especially "OPS Per Part," are extremely small and may not reflect meaningful operational prioritization unless scaled or contextualized

Revenue-Based OPS Adds Financial Context:

- Insulet Mech Rail and Insulet Battery Clip show higher scores under "Our OPS / Revenue," suggesting lower return efficiency per dollar earned, which may not show in Brunk's metrics.

Our approach for the analysis

Analyze ops per revenue and ops per part to identify industries and parts with higher scores and lower scores



Identification

Identify which industries/parts have high/low scores

Analysis



Analyze the trends and find patterns

Analysis BREAKDOWN



Conclude the reasoning behind trends

Conclusion

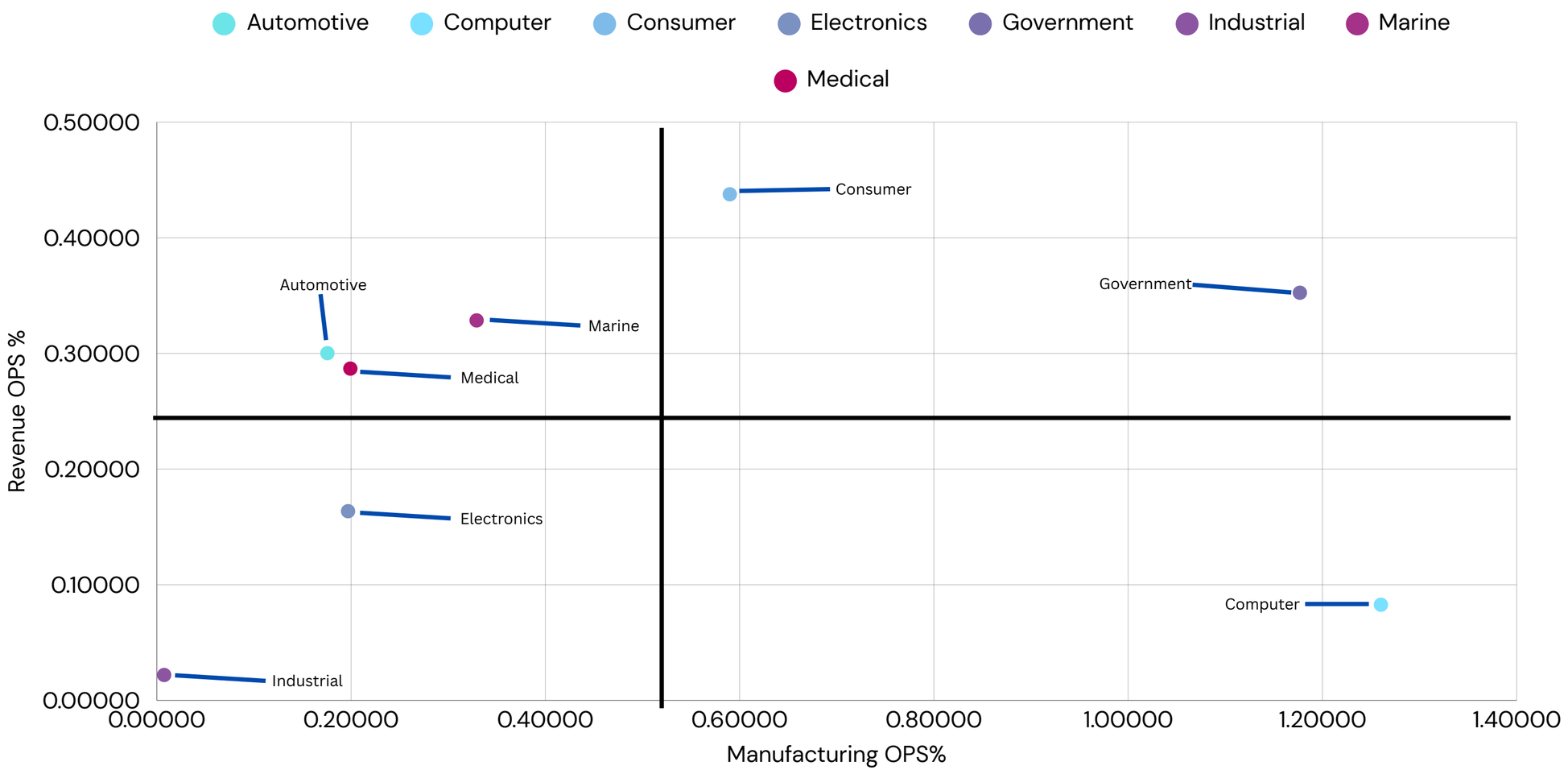
Recommend ways to improve the OPS Score



Recomendation

Plotting the Data

Graphing the median of each field and Identifying outliers



Star Industries

Industrial + Electronics

High efficiency, high revenue retention

Prioritize growing these industries — they are operationally and financially solid.

Scale them if market demand allows; they can absorb growth efficiently.

Study their processes to replicate success across weaker areas.

Cash Cow Industries

Automotive + Medical + Marine

High efficiency, consistent performers

Accounts for the majority of parts - the largest section and reliably brings in money

Investigate post-production losses

Maintain current production levels — manufacturing is not the issue.

Question Mark Industry

Computer

Low efficiency, high revenue retention

High sales prices make up for manufacturing problems

Potential star industry - revenue already comes in at a high rate

Target this area for lean initiatives — reduce rework, scrap, and waste.

Dog Industries

Consumer + Government

Low efficiency, low revenue retention

Contains problems in both manufacturing and revenue production

Consider scaling down or exiting these industries if no strategic benefit exists

Reallocate resources toward higher-performing areas unless justified otherwise

Risks

Heavily investing in Star Industries might inherently put less of an emphasis on Cash Cow Industries, which make up the majority of the parts made

Question Mark Industries (i.e. Computers) have the ability to turn into Stars but also have the ability to turn into Dogs

Placing less of an emphasis on Dog Industries might decrease Brunk's credibility (e.g. Government)

Next Steps

Investigate market expansion opportunities for Star Industries

Experiment with increases in pricing for parts in Cash Cow Industries to address high revenue OPS scores

Investigate OPS outliers to identify root causes of extreme inefficiency in production and revenue leakage

The Team leading the OPS Score Analysis

The Brunk Industries IX Team

Thank you!



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Outliers (Manufacturing)

Automotive:
•34R-15055-1 (COVER PLATE) -- Manufacturing OPS of 6.968090099

Computer:
•500827517T (RETENTION PLATE 18MM X 18MM) --Manufacturing OPS of 11.18952811

Marine:
•398-5454-A28 (LAMINATION ASSEMBLY) -- Manufacturing OPS of 48.96220752**
•16409A1 (LAMINATION ASSEMBLY) -- Manufacturing OPS of 48.10002694
•398-858404-A1 (LAMINATION ASSEMBLY) -- Manufacturing OPS of 22.64236777

Medical:
•AS00225 (TUBULAR SHAT ASSEMBLY – PROG WING ATC) – Manufacturing OPS of 15.07408578
•M955171A-001 (SHIELD BOTTOM (20ml)) – Manufacturing OPS of 38.52340758**

Outliers (Revenue)

Automotive:

•34R-15055-1 (COVER PLATE) – Revenue OPS of 4.432899976**

Marine:

•398-5454-A28 (LAMINATION ASSEMBLY) – Revenue OPS of 1.923497224**

Medical:

•383619 (ANTI-BACKUP STAMPING) – Revenue OPS score of 2.956159342

•10000165518 (OBI - INTERLOCK, HEMMED) – Revenue OPS score of 1.895193699

•A08034-001 (WELD PLATE, ICR ANTENNA CONNECTOR) – Revenue OPS of 1.693836868

•M20327P01 (LOCK LEVER) – Revenue OPS score of 1.596667405

•M936654A-001 (LEADFRAME) – Revenue OPS score of 2.395920881

•M955171A-001 (SHIELD, BOTTOM (20ml)) – Revenue OPS of 2.536873325**

Fixing Manufacturing To Improve Outlier OPS Scores

Decreasing Quality time by 50% means:

- Part 34R-15055-1 (COVER PLATE) - Manufacturing OPS decreases from 6.968090099 to 3.48438841,

Revenue OPS decreases from to 4.432899976 2.216668424

- 398-5454-A28 (LAMINATION ASSEMBLY) – Manufacturing OPS decreases from 48.96220752 to 26.08653262

Revenue OPS decreases from 1.923497224 to 1.024818439

- 398-858404-A1 (LAMINATION ASSEMBLY) -- Manufacturing OPS decreases from 22.64236777 to 16.62859261

Decreasing Scrap by 50% means:

- 16409A1 (LAMINATION ASSEMBLY) - Manufacturing OPS decreases from 48.10002694 to 27.53385563

Decreasing Scrap AND Quality Time by 50% means:

- AS00225 (TUBULAR SHAT ASSEMBLY – PROG WING ATC) – Manufacturing OPS decreases from 15.07408578 to 9.660321983
- M955171A-001 (SHIELD BOTTOM (20ml)) – Manufacturing OPS decreases from 38.52340758 to 33.48646318

Increasing Revenue To Improve Outlier OPS Scores

Some parts had a good production OPS score, but a poor revenue OPS score. Improving an already low manufacturing OPS score may be difficult, so for these we tested raising revenue. This can be achieved by price increases, heavier advertising for specific products, or however Brunk sees fit.

Increasing part revenue by 50% means:

- 383619 (ANTI-BACKUP STAMPING) – Revenue OPS decreases from 2.956159342 to 1.970772895
- 10000165518 (OBI - INTERLOCK, HEMMED) – Revenue OPS decreases from 1.895193699 to 1.263462466
- A08034-001 (WELD PLATE, ICR ANTENNA CONNECTOR) – Revenue OPS decreases from 1.693836868 to 1.129224578
- M20327P01 (LOCK LEVER) – Revenue OPS score decreases from 1.596667405 to 1.064444936
- M936654A-001 (LEADFRAME) – Revenue OPS score decreases from 2.395920881 to 1.597280587

Note: Changing these numbers does not result in a lower industry *median* than we calculated, but would certainly lower the industry *mean*.